19 VECTOR DERIVATIVES

19.1 Directional derivatives

Suppose that $\boldsymbol{\xi}$ and $\mathbf{u}$ are vectors in $\mathbb{R}^n$ and that $\mathbf{u}$ is of unit length (i.e. $\|\mathbf{u}\| = 1$). We shall think of $\boldsymbol{\xi}$ as determining a point in $\mathbb{R}^n$ and of $\mathbf{u}$ as determining a direction. The set of all vectors of the form

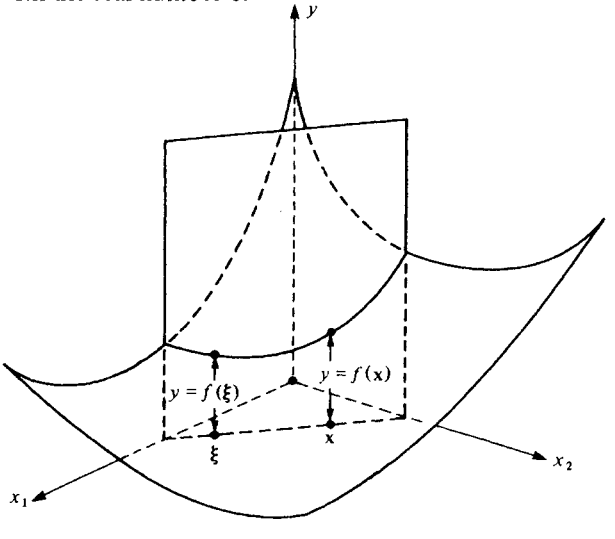
$$\mathbf{x} = \boldsymbol{\xi} + t\mathbf{u} \quad (t \in \mathbb{R})$$

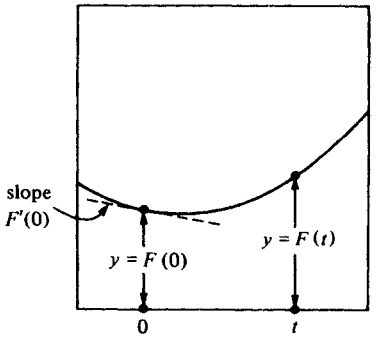
can then be thought of as the straight line through the point $\boldsymbol{\xi}$ in the direction of $\mathbf{u}$. The parameter t measures the distance from $\boldsymbol{\xi}$ to $\mathbf{x}$ along this line. (See §18.13.)

If $f\colon \mathbb{R}^n \to \mathbb{R}$, we can define a function $F\colon \mathbb{R} \to \mathbb{R}$ by

$$F(t) = f(\boldsymbol{\xi} + t\mathbf{u}).$$

In the case when $n = 2$, the geometry of the situation is very simple. One may regard $y = F(t)$ as the equation of the curve obtained by slicing the surface $y = f(\mathbf{x})$ with a vertical plane through the point $\boldsymbol{\xi}$ which is parallel to $\mathbf{u}$. The diagram below shows the surface $y = f(\mathbf{x})$ sliced by the appropriate vertical plane. The following diagram suppresses everything outside this vertical plane and relabels the point $\mathbf{x} = \boldsymbol{\xi} + t\mathbf{u}$ with the real number t. Correspondingly, $\boldsymbol{\xi}$ is relabelled with the real number 0.





It is apparent from these diagrams that $F'(0)$ is the slope (or rate of increase) of the surface $y = f(\mathbf{x})$ at the point $\boldsymbol{\xi}$ in the direction $\mathbf{u}$. We call this important quantity the *directional derivative* of f at the point $\boldsymbol{\xi}$ in the direction $\mathbf{u}$ and denote it by $D_{\mathbf{u}}f(\boldsymbol{\xi})$. Recalling the definition of a derivative given in §10.1, we have that

$$D_{\mathbf{u}}f(\boldsymbol{\xi}) = \lim_{t\to 0} \frac{f(\boldsymbol{\xi} + t\mathbf{u}) - f(\boldsymbol{\xi})}{t}.$$

The same notation is also used when f is a vector-valued function and $\mathbf{u}$ is not necessarily a unit vector. But then of course the preceding geometric interpretation does not apply.

19.2 Example Let $f\colon \mathbb{R}^2 \to \mathbb{R}$ be defined by $f(x_1, x_2) = x_1x_2$. Find the slope of the surface $y = f(x_1, x_2)$ at the point $(1, 2)$ in the direction $(3, 4)$.

The vector $(3, 4)$ is not a unit vector and so we begin by replacing it by the unit vector $(3/5, 4/5)$ which points in the same direction. We have to calculate

$$D_{\mathbf{u}}f(\boldsymbol{\xi}) = \lim_{t\to 0} \frac{f(\boldsymbol{\xi} + t\mathbf{u}) - f(\boldsymbol{\xi})}{t}$$

in the case when $\boldsymbol{\xi} = (1, 2)$ and $\mathbf{u} = (3/5, 4/5)$. We obtain that

$$D_{\mathbf{u}}f(\boldsymbol{\xi}) = \lim_{t\to 0} \frac{(1 + 3t/5)(2 + 4t/5) - (1)(2)}{t}$$

$$= \lim_{t\to 0} \left(2 + \frac{12t}{25}\right) = 2.$$

It follows that the slope of the surface $y = x_1x_2$ at the point $(1, 2)$ in the direction $(3, 4)$ is equal to 2. The diagram illustrates the curve $x_1x_2 = 2$ which is called a *contour* of the function f. Points on the surface $y = f(x_1, x_2)$ above this contour are all at the same height. The unbroken arrow drawn at $(1, 2)$ indicates